

DAM assignment week 12

Ina, Ida, Marie & Mark

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- Explore Homicide rates in NW Europe
 - Load the available data from ourworldindata.org
 - Data inspections
 - Individual countries
 - Compare the trends in homicide with the pattern of reign duration among Danish rulers through time.
 - Final question: are we more civilized today?

```
knitr::opts_chunk$set(echo = TRUE,  
  message=FALSE)
```

Explore Homicide rates in NW Europe

In this exercise, you will load a Manuel Eisner's dataset that captures the developments in homicides from Middle Ages to 20th century.

Your task: Explore the data and create visualisations as instructed in the script.

- Upload to Github this .Rmd file with solutions and its .html knitted version.
- Submit to Brightspace a document that follows the template and contains (1) URL of this script and its knitted version in Github, (2) one plot of your choice, and (3) answer of 250 words to the Final Question.

This exercise is based on the dataset provided by OurWorldInData project (<https://ourworldindata.org/homicides>) based at the Oxford University.

```
library(tidyverse)
```

Load the available data from ourworldindata.org

You should always interrogate the source of your data. Who compiled it, from where, what is missing, how representative the data are? Check the data/Metadata.txt to learn about the data provenance.

```
Western_Europe <- read_csv("data/homicide-rates-across-western-europe.csv", na = c("NULL",  
  ""))
```

Data inspections

general data inspection

```
head(Western_Europe)
```

```
## # A tibble: 6 × 4
##   Entity Code   Year Homicide rate in Europe over long-term (per 100,000) (ho...1
##   <chr>   <chr> <dbl>                                <dbl>
## 1 England <NA>   1300                                23
## 2 England <NA>   1550                                7
## 3 England <NA>   1625                                6
## 4 England <NA>   1675                                4
## 5 England <NA>   1725                                2
## 6 England <NA>   1775                                1
## # i abbreviated name:
## #   1Homicide rate in Europe over long-term (per 100,000) (homicides per 100,000 people)`
```

```
glimpse(Western_Europe)
```

```
## Rows: 206
## Columns: 4
## $ Entity                                <c
hr> ...
## $ Code                                  <c
hr> ...
## $ Year                                  <d
bl> ...
## $ `Homicide rate in Europe over long-term (per 100,000) (homicides per 100,000 people)` <d
bl> ...
```

Name changed to homicides_per_100k

```
# names() function produces the names of columns as a character vector with 4 elements
names(Western_Europe)
```

```
## [1] "Entity"
## [2] "Code"
## [3] "Year"
## [4] "Homicide rate in Europe over long-term (per 100,000) (homicides per 100,000 people)"
```

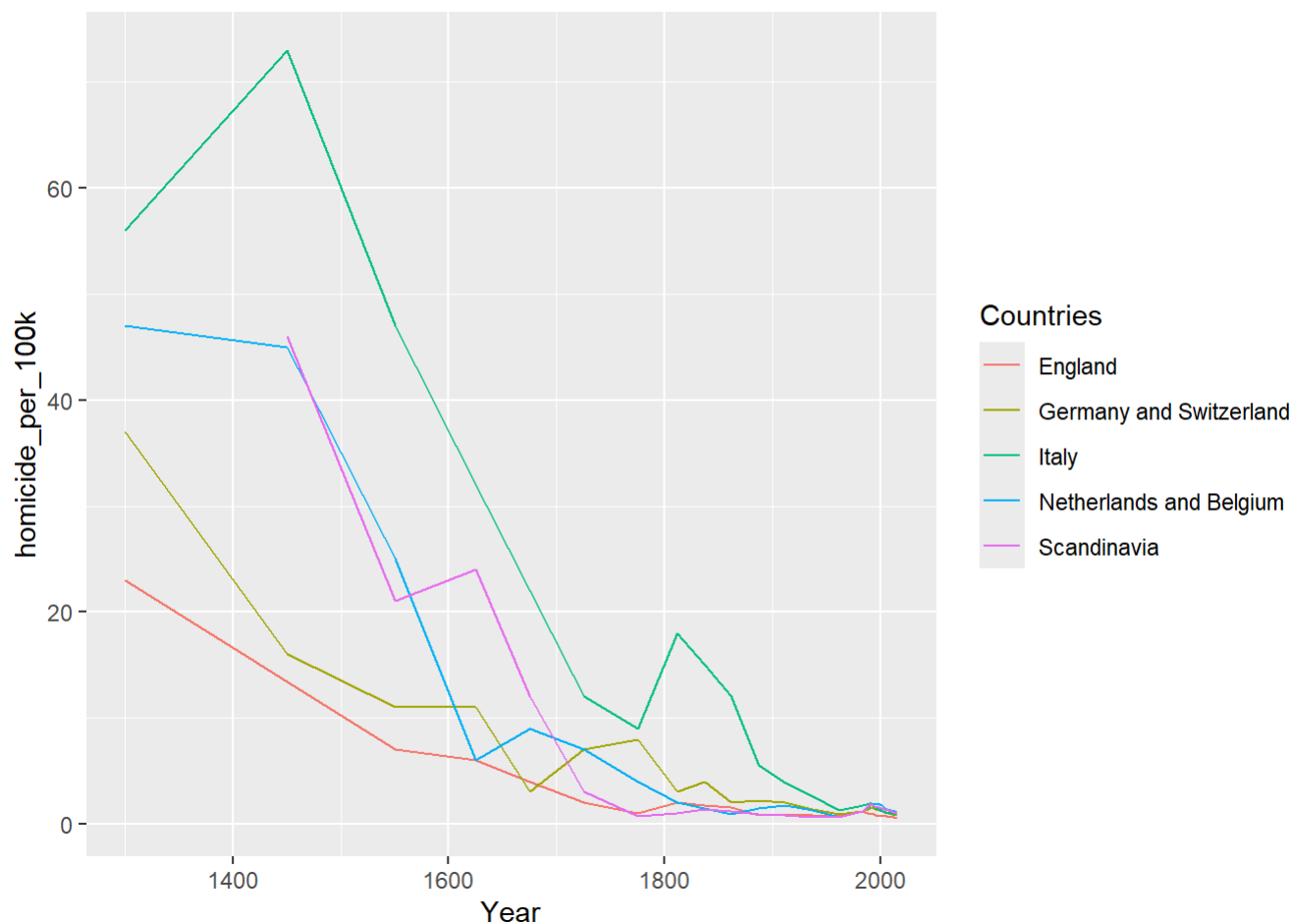
```
names(Western_Europe)[4] <- "homicide_per_100k"

names(Western_Europe)[1] <- "Countries"
```

##long-term trend in homicides

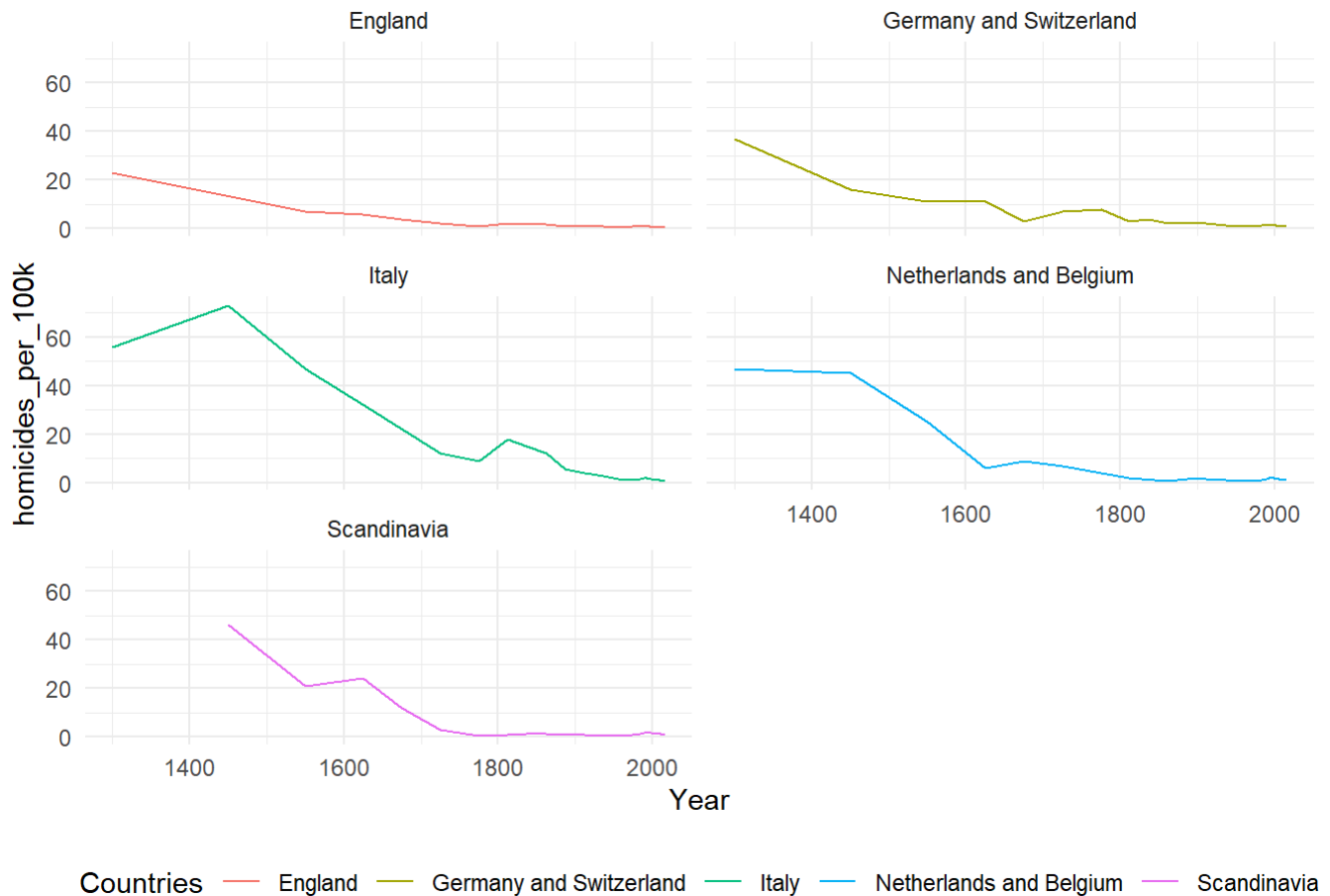
```
# Homicides over time for all countries

#The code below is an AES version
Western_Europe %>%
  ggplot(aes(x = Year, y = homicide_per_100k, colour = Countries)) +
  geom_line()
```



Individual countries

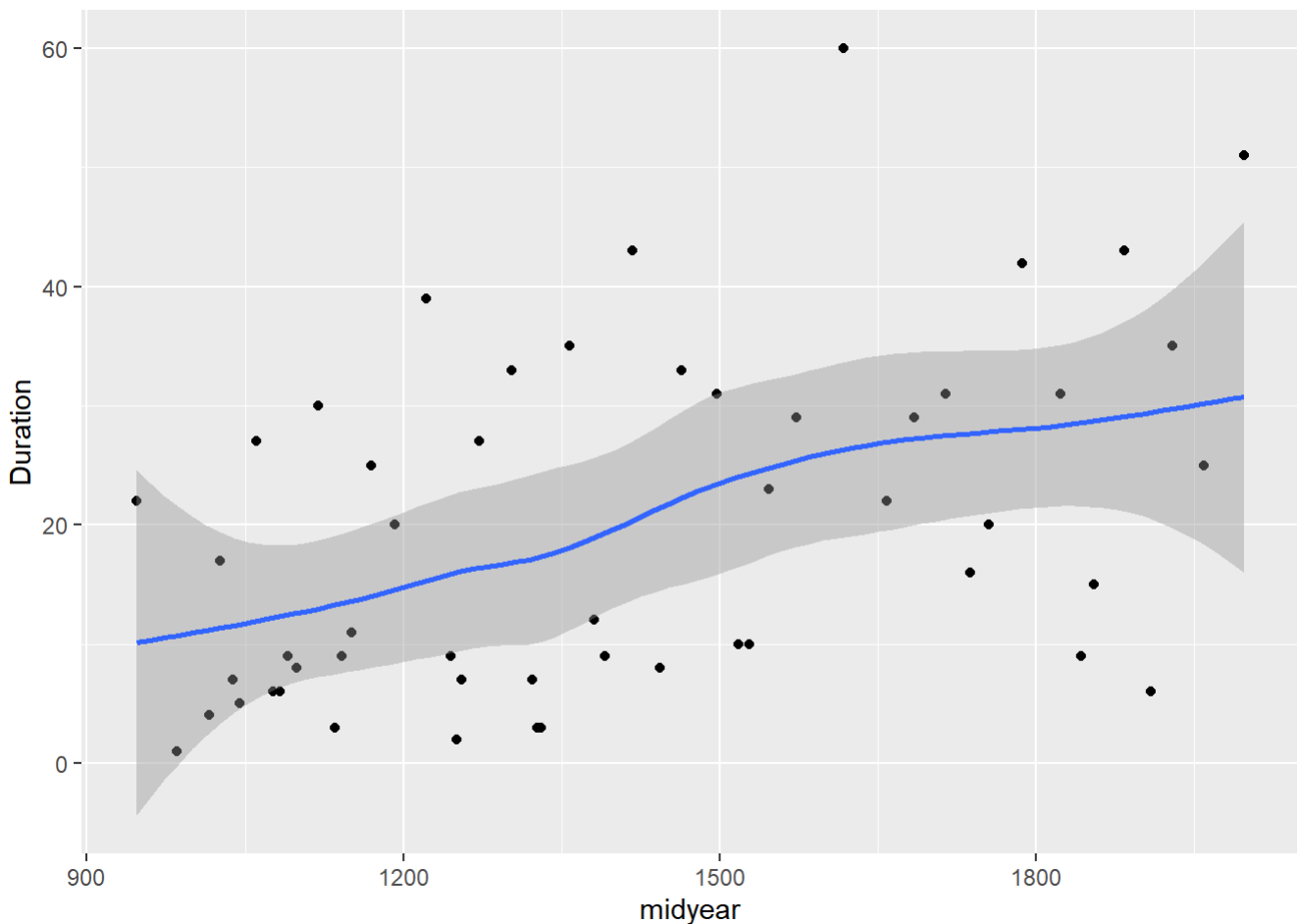
```
# Individual countries over time
Western_Europe %>%
  ggplot(aes(x = Year, y = homicide_per_100k, colour = Countries)) +
  geom_line() +
  facet_wrap(~ Countries, ncol = 2) +
  theme_minimal() +
  labs(y = "homicides_per_100k") +
  theme(legend.position = "bottom")
```



Compare the trends in homicide with the pattern of reign duration among Danish rulers through time.

```
# Kings data sequence
kings <- read_csv("data/kings.csv")
kings %>%
  mutate(
    Start_year = parse_number(Start_year),
    End_year   = parse_number(End_year),
    Duration   = End_year - Start_year,
    midyear    = (Start_year + End_year) / 2
  ) %>%
  filter(!is.na(Duration)) %>%
  ggplot(aes(x = midyear, y = Duration)) +
  geom_point() +
  geom_smooth()
```

```
## Warning: There were 2 warnings in `mutate()`.
## The first warning was:
## i In argument: `Start_year = parse_number(Start_year)`.
## Caused by warning:
## ! 2 parsing failures.
## row col expected actual
## 2 -- a number NULL
## 4 -- a number NULL
## i Run `dplyr::last_dplyr_warnings()` to see the 1 remaining warning.
```



We see an increase in general duration of rule for the monarchs, while the homicide rates fall. So duration over time rises in unison with homicides falling

##Final tasks

1. Plot: In the faceted plot above, move the legend from the current position on the side to below the facets, and label it "Country" instead of "Entity".
2. Rmarkdown:
 - edit the author of the document, and convert 'Final Tasks' into heading #2 (like the other headings)
 - add a floating table of contents to your Rmarkdown document,
 - provide informative chunk-names and edit flags in your R chunks to TRUE, and
 - automatically generate a timestamp to show when the document was last updated. (Hint: check the Rmarkdown episode in our Data Carpentry tutorial)

Final question: are we more civilized today?

Looking at the data we see a drastic decrease of homicides in all countries we look at, to understand the question, you have to look at human evolution. 500 years ago there was, for example, not a doubt that god existed, and wars were fought over him, however today some people see it differently. You cant compare the mentality of people 500 years ago, to today. Today we live in a completely different world than we did 500 years ago, the human mentality has changed, and therefore become more civilized than we were. Killing does still occur, but nowhere near the scale we see earlier in our history. Laws, mentality and science has changed, we have the ability to end the world, which could be what keeps us from matching the numbers of homicides we had 500 years ago.